MIL-R-43040B

2 March 1972

SUPERSEDING
MIL-R-43040A
30 September 1968

MILITARY SPECIFICATION

# REMOVAL TOOLS, CONNECTOR ELECTRICAL CONTACT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

* 1.1 Scope.- This specification covers removal tools for electrical equipment (see 6.1).

* 1.2 Classification.- The removal tools shall be of the following types and sizes, as specified (see 6.2):

Type I - Remover, Connector Contact

Size - 0.063 - inch bit point diameter
Size - 0.120 - inch bit point diameter
Size - 0.187 - inch bit point diameter

Type II - Extractor, Turnlock Fastener

## * 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-A-225/3 - Aluminum Alloy Bar, Rod, and Wire; Rolled, Drawn, or Cold Finished, 2011

FSC 5120

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QQ-S-637 - Steel Bar, Carbon, Cold Finished (Standard Quality, Free Machining).
QQ-T-580 - Tool Steel, Carbon and Carbon-Vanadium.
QQ-W-412 - Wire, Steel, Alloy (General Purpose Quality), for Mechanical Springs.
VV-L-800 - Lubricating Oil, General Purpose, Preservative (Water Displacing Low Temperature).
GGG-B-1222 - Bit, Screwdriver and Adapter, Screwdriver Bit.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Box, Setup.

MILITARY

MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-S-8503 - Steel Bars, Chrome-Vanadium (6150).
MIL-A-8625 - Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-C-13924 - Coating, Oxide, Black, for Ferrous Metals.
MIL-L-21260 - Lubricating Oil, Internal Combustion Engine, Preservative.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MS-171471 - Pin, Spring, CRES.

* DRAWINGS

ARMY NATICK LABORATORIES

12-1-102 - Removal Tools, Electrical Equipment, Type I Remover, Connector Contact; Assemblies and Details.
12-1-175 - Removal Tool, Electrical Equipment, Type II, Extractor Assembly and Details.

(Miniature copies of Drawings 12-1-102 and 12-1-175, identified as figures 1 and 2 are attached for information purposes only).

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).



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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

American Society for Testing and Materials

E18-67 - Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington D.C. 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

American National Standards Institute, Incorporated

American National Standard B46.1-62 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, N.Y. 10018.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

* 3.1 First article.- When specified, the contractor shall furnish a sample for first article inspection and approval (see 4.2, 6.2 and 6.5).



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* 3.2 Materials.- Materials shall conform to the referenced specifications listed on the applicable drawings in 2.1 and requirements specified herein.

* 3.3 Construction.-

3.3.1 Type I and II tools.- The type I and II removal tools shall conform to the design, details, and dimensions as shown on applicable drawings and to the requirements specified herein.

* 3.3.1.1 Remover bits.- The bits shall have a Rockwell hardness between C43 to C48 when tested as specified in 4.4.1. The surface finish shall be no coarser than 125 micro-inches in accordance with ANSI B46.1-62.

* 3.3.1.2 Bit, screwdriver.- The screwdriver bit shall conform to type II, class 4, 1/4 inch width of GGG-B-1222.

* 3.4 Finish.-

* 3.4.1 Plating.- Plating shall be smooth, adherent, free of flaking and imbedded foreign matter, and areas of no film.

* 3.4.1.1 Set screws, types I and II.- The type I and type II removal tool set screws shall be cadmium plated in accordance with the manufacturer's commercial practice.

* 3.4.1.2 Stud, type II.- The type II removal stud shall be nickel or chromium plated in accordance with the manufacturer's commercial practice.

* 3.4.1.3 Handles.- The handles of type I and II removal tools shall be given an anodic coating conforming to types and classes of MIL-A-8625 as follows:

(1) Type I, class 1 for type I removal tool.
(2) Type II, class 2, black for type II removal tool.

* 3.4.1.4 Bits.- Bits of type I and II removal tools shall be given a black oxide coating conforming to class 1 of MIL-C-13924.

* 3.5 Marking for identification.- Type I and II removal tools shall be marked for identification as shown on Drawings 2-1-102 and 12-1-175, as applicable. The markings shall be indented on the handles.



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* 3.6 Workmanship.- Type I and II removal tools shall be free from burrs, slivers, rough and sharp edges. Components shall not be loose, fractured, broken or distorted.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 First article inspection.- When a first article is required, it shall be examined for the defects specified in table I, dimensions specified, and tested in accordance with 4.4.1. The presence of any visual defects, dimensions, not within specified requirements or failure to pass the test shall be cause for rejection of the first article.

4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

* 4.3.1.1 Intermediate test.- Prior to assembly, the bits of type I and II removal tools shall be tested as specified in 4.4.1 for compliance with hardness requirements of 3.3.1.1 and 3.3.2.1, as applicable. Failure to pass the test shall be considered a defect. The lot shall be all the same size bits of type I removal tools offered for inspection at one time. The sample unit shall be one bit of the type I removal tool. The inspection level shall be II with an acceptable quality level (AQL) of 1.5 expressed as defects per hundred units.

4.3.2 End item inspection.- The lot shall be all type I removal tools with specified size tool bit and all type II removable tools offered for inspection at one time. The sample unit shall be one complete specified type I removal tool with assembled bit or one complete type II removal tool.



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4.3.2.1 Visual examination.- The end item shall be examined for the defects in table I. The inspection level shall be II and the AQL shall be 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I.- Classification of defects

| Examine | Defect | Classification Major | Minor |
|---|---|---|---|
| Finish | Not as specified. | X | |
| | Not smooth, adherent and free of flaking and imbedded foreign matter or areas of no film. | X | |
| | Handles not anodic coated. | X | |
| | Oxide coat not black when applicable. | X | |
| Workmanship (General) (Components) | Not free of burrs, slivers, rough and sharp edges. | | X |
| | Components loose, fractured, broken or distorted. | X | |
| Marking | Missing, not legible, incorrect, not specified type or not located as specified. | | X |

4.3.2.2 Dimensional examination.- Examination shall be made of the removal tools for dimensional requirements specified. Any dimensions not within specified requirements shall be classified as a defect. The inspection level shall be S-2 with an AQL of 2.5, expressed in terms of defects per hundred units.

4.3.3 Examination of preparation for delivery.- An examination shall be made to determine that packaging, packing and marking as required by section 5 are complied with. Defects shall be as indicated in table II. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be all of the containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.



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TABLE II.- Examination of preparation for delivery

| Examine | Defect |
|---|---|
| Marking, (exterior and interior) | Omitted; incorrect; illegible; of improper size, location, sequence, or method of application. |
| Workmanship | Inadequate application of components, such as inadequate closure of interior boxes, or bulging or distortion of containers. |
| Weight | Weight of contents per container is greater than specified. |

4.3.3.1 Examination of shipping containers.- When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure and strapping shall be in accordance with the appendix of that specification.

4.4 Test.-

* 4.4.1 Hardness test.- Hardness test for type I removal tool bits shall be conducted in accordance with ASTM Test Method E18-67. The hardness shall be based on the average of three determinations as follows:

The hardness shall be obtained on each bit of one size, at each end and at the entry of the bit. Any bit having a hardness less than RC43 or more than RC48 shall constitute failure of this test (see 3.3.1.1).

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.- The bit or stud of each removal tool shall be thoroughly cleaned and dried, and coated with a preservative oil conforming to VV-L-800, or grade 1, 2 or 3 of MIL-L-21260. Ten removal tools of one type and size only shall then be wrapped in barrier material conforming to type II, grade A, class 1 or 2 of MIL-B-121, and packaged in a snug-fitting folding or setup paperboard box conforming to variety 1, style III or VIII, type G, class 1 of PPP-B-566; or type I, variety 1, class A, style optional of PPP-B-676, respectively. Closure shall be in accordance with the appendix of the applicable box specification.

5.1.2 Level C.- Removal tools shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the



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first receiving activity. The supplier may use his standard practice when it meets this requirement.

* 5.2 Packing.- Packing shall be level A, B or C as specified (see 6.2).

* 5.2.1 Level A.- Removal tools of one size only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. The weight of contents of each shipping container shall not exceed the limitations of the container specification. Closure, waterproofing by means of tape, and reinforcing shall be in accordance with the appendix of PPP-B-636.

* 5.2.2 Level B.- Removal tools of one type and size only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF, variety SW, or type SF, class domestic of PPP-B-636. The weight of contents of each shipping container shall not exceed the limitations of the container specification. Closure shall be in accordance with method II of the appendix of PPP-B-636.

* 5.2.2.1 When specified (see 6.2), the shipping cotainer shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix thereto.

* 5.2.3 Level C.- Removal tools, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

* 5.3 Marking.- In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

* 6.1 Intended use.- The removal tools covered by this specification are used to remove contacts from electrical connectors, and latch mechanisms of printed circuit boards of electrical equipment.

* 6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and size required (see 1.2).



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(c) When a first article is required (see 3.1).
(d) Selection of applicable levels of preservation, packaging, and packing (see 5.1 and 5.2).
(e) When weather-resistant fiberboard shipping containers are required for level B packing (see 5.2.2.1).

* 6.3 The type I removal tools covered by this specification are components of Tool Kit, Electrical Connector, FSN 5180-708-3423, which is a component of:

(a) Tool Kit: Automotive Maintenance, Organization Maintenance Common Set No. 1, FSN 4910-754-0654
(b) Shop Set, Fuel and Electrical System, Field Maintenance, Basic, FSN 4910-754-0714
(c) Shop Set, Automotive Repair: Field Maintenance, Basic, FSN 4910-754-0705

6.4 The type II removal tool, Extractor, Turnlock Fastener carries a FSN of 5120-785-6052.

* 6.5 First article.- When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample consisting of a complete type I or type II removal tool. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.6 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.



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Custodians:

Army - GL
Navy - SH
Air Force - 84

Review activity:

Army - WC

User activities:

Army - EL, AT, ME, MI
Navy - AS, OS, YD

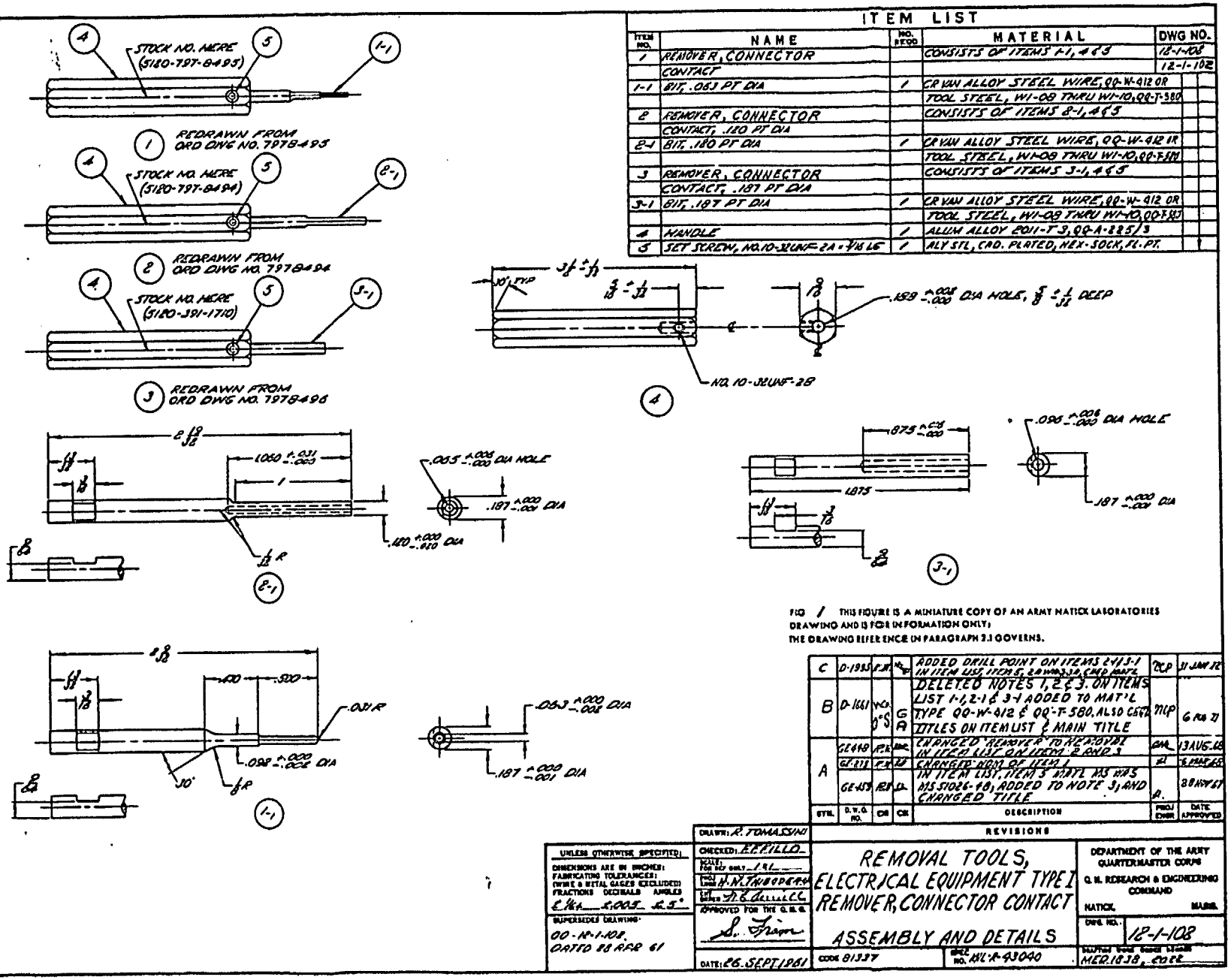
Preparing activity:

Army - GL

Project No. 5120-0667



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## ITEM LIST

| ITEM NO. | NAME | NO. REQD | MATERIAL | DWG NO. |
|---|---|---|---|---|
| 1 | REMOVER, CONNECTOR | | CONSISTS OF ITEMS 1-1, 4 & 5 | 12-1-108 |
| | CONTACT | | | 12-1-102 |
| 1-1 | BIT, .063 PT DIA | 1 | CR VAN ALLOY STEEL WIRE, QQ-W-412 OR | |
| | | | TOOL STEEL, W1-08 THRU W1-10, QQ-T-580 | |
| 2 | REMOVER, CONNECTOR | | CONSISTS OF ITEMS 2-1, 4 & 5 | |
| | CONTACT, .120 PT DIA | | | |
| 2-1 | BIT, .180 PT DIA | 1 | CR VAN ALLOY STEEL WIRE, QQ-W-412 OR | |
| | | | TOOL STEEL, W1-08 THRU W1-10, QQ-T-580 | |
| 3 | REMOVER, CONNECTOR | | CONSISTS OF ITEMS 3-1, 4 & 5 | |
| | CONTACT, .187 PT DIA | | | |
| 3-1 | BIT, .187 PT DIA | 1 | CR VAN ALLOY STEEL WIRE, QQ-W-412 OR | |
| | | | TOOL STEEL, W1-08 THRU W1-10, QQ-T-580 | |
| 4 | HANDLE | 1 | ALUM ALLOY 2011-T3, QQ-A-225/3 | |
| 5 | SET SCREW, NO.10-32UNF-2A × 7/16 LG | 1 | ALY STL, CAD. PLATED, HEX-SOCK, FL.PT. | |

FIG 1 THIS FIGURE IS A MINIATURE COPY OF AN ARMY NATICK LABORATORIES DRAWING AND IS FOR INFORMATION ONLY; THE DRAWING REFERENCE IN PARAGRAPH 3.1 GOVERNS.

| STYL | D.W.G. NO. | CH | CK | DESCRIPTION | PROJ ENGR | DATE APPROVED |
|---|---|---|---|---|---|---|
| C | D-1933 | F.M. | | ADDED DRILL POINT ON ITEMS 2-1 & 3-1 IN ITEM LIST, ITEM 5, 28 NF-3 2A CMD MATL | TCP | 21 JAN 58 |
| B | D-1661 | YOU | G A | DELETED NOTES 1, 2 & 3. ON ITEMS LIST 1-1, 2-1 & 3-1 ADDED TO MAT'L TYPE QQ-W-412 & QQ-T-580. ALSO CHGD TITLES ON ITEM LIST & MAIN TITLE | TCP | 6 FEB 57 |
| A | GE 449 | A.E. | [illegible] | CHANGED "REMOVER" TO "REMOVAL" IN ITEM LIST & ON ITEMS 2 AND 3 | GM | 13 AUG 56 |
| | GE-477 | F.M. | [illegible] | CHANGED SIZE OF ITEM 1 | GM | 6 MAR 57 |
| | GE-457 | R.B. | G. | IN ITEM LIST, ITEM 5 MATL AS PER MS 51026-98, ADDED TO NOTE 3, AND CHANGED TITLE | A. | 28 MAY 57 |

REVISIONS

DRAWN: R. TOMASSINO
CHECKED: R. P. FILLO
SCALE: FOR REF ONLY - 1/1
PROJ. ENGR: H. N. THROOPER
LIT. ENGR: H. E. DELLACC
APPROVED FOR THE O.M.R.&D.: S. Shapiro
DATE: 26 SEPT 1961

UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN INCHES; FABRICATING TOLERANCES: (WIRE & METAL GAGES EXCLUDED) FRACTIONS DECIMALS ANGLES ± 1/64 ± .005 ± .5°

SUPERSEDES DRAWING: 00-M-1-102, DATED 28 APR 61

# REMOVAL TOOLS, ELECTRICAL EQUIPMENT TYPE I REMOVER, CONNECTOR CONTACT ASSEMBLY AND DETAILS

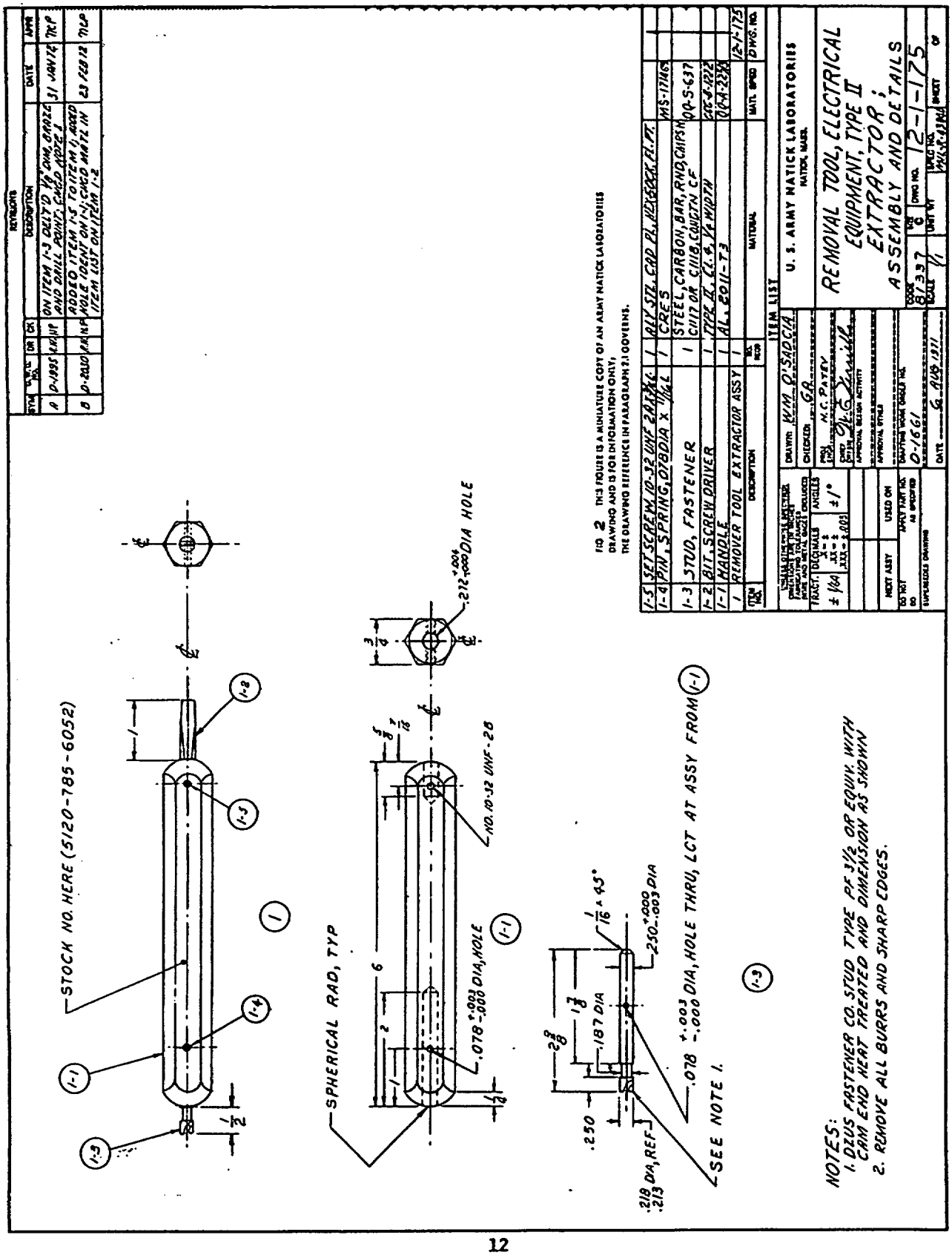
DEPARTMENT OF THE ARMY
QUARTERMASTER CORPS
O. M. RESEARCH & ENGINEERING COMMAND
NATICK, MASS.

DWG. NO. 12-1-108

CODE 81337
SIZE NO. MIL-R-43040
MED 1838, 2062

MIL-R-43040B

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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

**INSTRUCTIONS:** This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION: Removal Tools, Connector Electrical Contact MIL-R-43040B

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

**DD FORM 1 JAN 66 1426** REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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